Joseph Bennett Zeru

https://worldofjoseup.live/resume
https://worldofjoseup.live/coding-projects

Education

Bachelor of Science, Computer Science, Mathematics May 2028
Rose-Hulman Institute of Technology Terre Haute, IN
Certificates: Quantum Science, Computing, & Engineering
Minors: Imaging, Artificial Intelligence

Skills

Relevant Courses: Deep Learning, Machine Learning, Linear Algebra I, Differential Equations II, Designing & Building AI Systems, Data Structures & Algorithms, Proofs, Combinatorics, Probability with Applied Statistics, Web Programming, Digital Systems
Programming Languages: Qiskit, Angular, Python, C++, C, Java, ARM Assembly, Verilog
Libraries/Tools: PyTorch/TensorFlow, Git, Firebase, PowerShell

Project Experience:

Remote Security Analysis Using a Neural Network
In-Progress Machine Learning Project *May 2026 - Present*

- Using a neural network to observe possible security risks for a repurposed 2014 Mac Mini being used as a NAS
- Fine-tuning a Security-Informed Neural Network (SINN) to check for suspicious remote access activity
- Building a responsive Angular front-end for local ease of use, as well as demonstrations to restrict NAS access

Budget-Management Tool
Web-App Personal Project *April 2026*

- Used DOM Manipulation to add/remove items dynamically
- Saved browser data using localStorage to persist budget items across sessions via local devices
- Used disability friendly font for easy access to users with visual impairments

Work Experience:

BinaryHeart Terre Haute, IN
Project Lead and Rose-Hulman Chapter Co-Founder *June 2025 – Present*

- Lead logistical projects on computer distribution across the greater Terre Haute area and Africa
- Develop technical projects that provide the public with easy-to-use automation tools
- Promote initiatives with executive members through advertising and marketing strategies across campus to promote the club and improve its growth

BMS Software Team Member Terre Haute, IN
BattChallenge RHIT – Vigo Voltage *September 2025 – January 2026*

- Utilized MATLAB and Simulink to code test condition optimizations for efficient battery systems in motor vehicles
- Developed software requirements to improve workflow by 30% of car battery subsystems such as plug-in charging
- Organized and cleaned BattLab cleanup to help improve workflow and efficiency across multiple teams

Leadership and Activities:

Vice President Terre Haute, IN
BinaryHeart @ RHIT, 501(c)(3) *September 2025 – Present*

- Taught students with little to no background in computer maintenance, how to repair out-of-life computers
- Donated over $3000 worth of refurbished computers globally from greater Terre Haute to universities in West Africa
- Ensure stable management of club budget through yearly auditing of the Treasurer

Secretary
Rose Fashion Club *April 2026 – Present*

CSSE Mentor
Rose-Hulman CSSE Mentor/Mentee Program *September 2025 – Present*

National & Regional Chapter Representative
National Society of Black Engineers *September 2024 – Present*